

The Null Token Knows: Reducing Message-Free Hallucination in ASR and NMT

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Abstract

Modern encoder-decoder systems can produce fluent text even when their input contains no recoverable message. We study this failure in ASR and NMT through the models’ reserved null tokens, asking whether the score for ending generation already carries a usable abstention signal. Across speech recognizers and translation models, we audit native null-token scores and scalar logit shifts. In Whisper, we additionally probe decoder states and compare supervised row edits with conventional external gates. The evaluated models often expose a useful abstention signal, but stock decoding does not reliably act on it. Raising the null-token score can sharply suppress fabrication, but aggressive intervention also deletes valid speech or shortens legitimate translations. These findings turn the null token into a diagnostic lens on hallucination and motivate evaluating abstention methods by both suppression and deletion costs, rather than by hallucination reduction alone.

1 Introduction

Attention-based encoder-decoder (AED) systems can fabricate fluent output from inputs that carry no message. Whisper (Radford et al. 2023) transcribes confident sentences from silence, room tone, and music, with documented downstream harms (Koencke et al. 2024; Frieske and Shi 2024); neural translators can likewise return fluent text for degenerate, message-free sources (Lee et al. 2018; Guerreiro, Voita, and Martins 2023; Raunak, Menezes, and Junczys-Dowmunt 2021). This failure does not reliably disappear with scale: across six Whisper checkpoints, our developmental endpoint is non-monotonic even as WER falls (Table 13) (Viakhirev, Borodin, and Mkrtchian 2026).

Operational systems counter this failure with native no-speech scores, VAD/endpointing, and reject options (Radford et al. 2023; FireRed Team 2026; Chow 1970). Other work re-trains attention heads (Wang et al. 2025), steers encoder features through sparse autoencoders (Aparin et al. 2026b), or distills against a clean teacher (Tripathi et al. 2025). Yet each evaluated family already exposes a reserved output coordinate: EOT in AED decoders, EOS in translation, and the semantically different alignment blank in CTC/RNN-T (Graves et al. 2006; Graves 2012). Because Whisper encoders also represent non-speech events (Gong et al. 2023), we ask a

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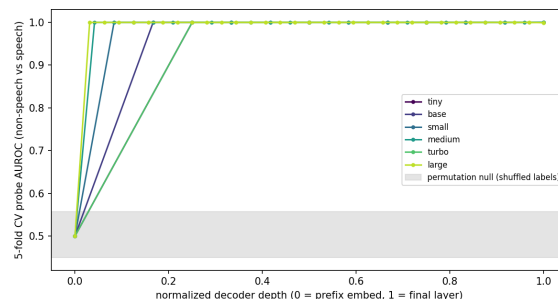


Figure 1: Developmental clip-stratified probes separate non-speech from speech after the first Whisper decoder block. The layer-0 embedding is at chance; the gray band is the permutation baseline. Folds are not grouped by source corpus, so this does not establish source-independent encoding.

narrower question: what battery-specific separation appears in decoder states and native scores, and what fabrication-deletion tradeoff follows from editing the reserved coordinate?

On the developmental battery, frozen decoder states linearly separate non-speech from speech after the first decoder block (Figure 1), and native null-token scores distinguish the two conditions even when the null token loses the output decision. We therefore use readout edits only as instruments for moving the abstention operating point. A scalar bias shifts constant prior odds; the analytic graft and trained row are supervised, state-dependent variants of one output coordinate. Their results do not establish an advantage over a scalar or external gate. Prior shift is a conditional model for the scalar case (Saerens, Latinne, and Decaestecker 2002; Lipton, Wang, and Smola 2018); we test where its operating-point predictions hold.

Our evidence comes from a developmental 1060-clip stress battery, a source-disjoint locked Whisper-small 300/300 evaluation, and a 15-model zoo spanning AED recognition, CTC/RNN-T, and neural translation. Three questions guide the analysis.

- **RQ1 (battery separation and readout).** How do decoder states and native null-token scores separate the developmental conditions across layers and positions (Section 4)?
- **RQ2 (checkpoint audit).** How do scalar and row edits

trade empty-target output against speech deletion, how do frozen checkpoints rank on the locked sensitivity grid, and where does the conditional prior-shift account fail (Section 5)?

- **RQ3 (boundary tests).** Which findings fail to transfer across architectures, modalities, languages, and long-form audio, and when are standard gates or segmentation safer (Sections 6 and 7)?

Measurement boundary. We measure deterministic *normalized lexical output* on spans with empty lexical targets. After removing documented non-lexical event tags, punctuation, and digits, any retained letter is fabricated by construction. The empty-target label is exact, but the score is normalizer-dependent and is not semantic adjudication. We exclude fabrication interleaved with genuine content, where the target is non-empty and no clean detector exists. WER cannot distinguish fabricated from misrecognized words, just as reference-overlap metrics miss generation faithfulness (Maynez et al. 2020; Ji et al. 2023). Although documented harms include interleaved fabrication (Koencke et al. 2024), our claims cover only whole-span empty lexical targets.

2 Background and Related Work

ASR and NMT hallucination is well documented (Koencke et al. 2024; Barański et al. 2025; Lee et al. 2018; Guerreiro, Voita, and Martins 2023). Native no-speech scoring and VAD/endpointing are the closest operational baselines (Radford et al. 2023; FireRed Team 2026); selective prediction formalizes the same reject option (Chow 1970; El-Yaniv and Wiener 2010). Complementary Whisper mitigations edit attention heads, encoder features, teachers, or emitted text (Wang et al. 2025; Aparin et al. 2026b,a; Tripathi et al. 2025). Whisper representations are already known to encode non-speech events (Gong et al. 2023; Viakhirev, Borodin, and Mkrtchian 2026); our distinct question is whether the decoder’s reserved null-token pathway carries a useful signal.

Under label shift, the Bayes decision changes by a closed-form prior-odds term (Saerens, Latinne, and Decaestecker 2002; Lipton, Wang, and Smola 2018). NMT models are miscalibrated, with EOS especially affected (Kumar and Sarawagi 2019); inference miscalibration also persists under generated context (Wang et al. 2020). We combine those results with frozen-state probes and logit-lens readouts (Alain and Bengio 2017; Belinkov 2022; Belrose et al. 2023) to characterize the available abstention signal. The intervention uses an in-vocabulary reject option (Chow 1970; El-Yaniv and Wiener 2010): a one-coordinate vector-scaling calibrator, whereas temperature scaling cannot change a greedy argmax (Guo et al. 2017).

Translation alternatives alter search with coverage, length, or lexical constraints (Wu et al. 2016; Murray and Chiang 2018; Post and Vilar 2018). Our edit changes the null-token scoring function. These alternatives are related inference-time methods, not matched baselines in our experiments. We test greedy ASR and each NMT backend’s fixed decoder; beam-width invariance and continuous long-form inference are outside the claim.

The established ingredients are rejection, label-shift correction, EOS calibration, native no-speech scoring, and VAD. Our empirical contribution is a diagnostic and falsification audit: it measures what direct null-coordinate edits can control, records their post-freeze fabrication–deletion cost among five Whisper-small checkpoint conditions, and identifies where stronger representation, prior-shift, risk-frontier, and cross-family hypotheses fail. Native and external rejectors are compared only on inspected developmental data. We do not claim to discover abstention or one shared mechanism for EOT, EOS, and CTC/RNN-T blank.

3 Experimental Setup

Data and metric. Our stress battery contains 630 non-speech and 430 speech clips (3.1 h total): synthetic silence/room tone; MUSAN noise/music (Snyder, Chen, and Povey 2015); ESC-50 (Piczak 2015); UrbanSound8K (Salamon, Jacoby, and Bello 2014); and clean/degraded FLEURS speech (Conneau et al. 2023). A deterministic filename-based assignment partitions MUSAN into disjoint calibration and evaluation halves. FLEURS test and LibriSpeech test-clean (Panayotov et al. 2015) are used only for evaluation (supplementary Table 12). For empty-reference clips, the endpoint is non-empty lexical output after the documented normalizer removes event tags, punctuation, and digits. It is deterministic given that normalizer, but does not count tag-only, number-only, or wrong-script output hidden by normalization. We call its clip incidence the *normalized lexical-output rate* (LOR), reserving “hallucination” for the broader failure class. Headline rates have clip-level Wilson intervals (Wilson 1927); paired `eot_row` effects use exact McNemar tests (McNemar 1947) with Holm correction (supplementary Tables 15–16). These finite-battery summaries are not cluster-robust or estimates of source-level transfer.

Models and interventions. The core zoo has six Whisper scales (Radford et al. 2023), Canary-1B (Puvvada et al. 2024), OWSM v3.1 (Peng et al. 2024), five CTC and two RNN-T recognizers, plus NLLB-200 and MarianMT (supplementary Table 13). Additional stock and readout checks cover Distil-Whisper, SeamlessM4T-v2-large, and Parakeet-TDT-1.1B. Their model-specific endpoints are reported in the supplement; no intervention claim is made for these checks.

Every stock/patch pair uses the same decoder. We distinguish three interventions. A scalar null-token bias adds a constant to one logit. An analytic row graft fits a supervised linear direction from labeled hidden states and adds it to one unembedding row. A trained EOT row updates the parameters of one output coordinate on a balanced calibration mix.

The supplement gives the common optimization recipe, regime matrix, run counts, data partitions, and pipeline checks. It also inventories the code, manifests, configurations, learned rows and biases, selection logs, and per-example outputs prepared for release.

Analysis and selection pools. The stress battery and reduced subsets were inspected during method development, so their intervals quantify within-set clip variation rather

than method adaptivity. After freezing methods and configurations, we evaluated once on 300 non-speech and 300 speech clips whose audio hashes, source recordings, and speakers do not overlap fitting or selection pools. The non-speech side draws from an unused silence seed and unused MUSAN, ESC-50, UrbanSound8K, and music recordings; speech covers clean, low-SNR, WHAM!-degraded, and accented inputs. A collision checker rejects a deliberately contaminated manifest. Test-clean had appeared previously as a reported benchmark, so the speech side is source-disjoint but not “virgin”; the unavailable spontaneous stratum is recorded as absent. Source disjointness addresses overlap, not deployment distribution shift.

We compare methods in two ways. On the developmental common-cost set, an operating point is reported as within an illustrative cap when at most 10% of real-speech clips are emptied; this threshold is not an application-derived utility. On the locked set, the pre-specified sensitivity family, frozen before decoding, is

$$C_\kappa = \text{fabricated lexical words/min} \\ + \kappa \text{ deleted reference words/min,} \\ \kappa \in \{0.5, 1, 2, 5, 10\}.$$

The weight κ combines relative exposure/prevalence and per-word harm; C_κ is a conditional-rate sensitivity index, not total WER or an expected deployment cost. The locked Whisper-small evaluation is the only post-freeze reporting pool; all zoo, probe, NMT, multilingual, and long-form results are developmental or boundary-setting. The full data-role ledger and source audit are in supplementary Table 5 and Section A.

4 RQ1: Developmental Separation and Readout

RQ1 measures how decoder states and native output scores separate the assembled developmental conditions. It also examines whether a decoding rule or later sequence positions account for the observed output.

Linear Recoverability in Decoder States

For each decoder layer ℓ we take the hidden state at the first content position (after the forced prefix) on the 1060-clip stress battery, and fit a logistic probe with 5-fold cross-validation to separate non-speech from speech. Layer 0 is the prefix-token embedding before any decoder block, and its probe is at chance. From the first decoder block onward, the clip-stratified probe reaches the maximum observed AUROC on all six scales (Figure 1; Table 1, first column).

A permutation null (shuffled labels) sits at 0.50 and a top-50 PCA probe reproduces the result, reducing concern about high-dimensional overfit. These probes establish linear recoverability on the assembled battery. Because folds are not grouped by source corpus, they do not by themselves establish a source-independent representation or a causal failure location.

The supervised probe measures in-battery linear recoverability. A complementary logit lens (nostalgebraist 2020;

Model	state AUROC@0			row/ns geometry		graft
	probe	ns	EOT	$\cos(\Delta, w_{\text{ns}})$	random p_{95}	
tiny	1.00	0.953	0.977	+0.119	0.103	fails
base	1.00	0.963	0.840	+0.043	0.092	fails
small	1.00	0.878	0.835	+0.114	0.112	fails
medium	1.00	0.915	0.728	+0.034	0.026	fails
turbo	1.00	0.988	0.995	−0.014	0.046	fails
large-v3	1.00	0.814	0.999	−0.032	0.046	fails

Table 1: Developmental Whisper state-probe and native-logit AUROC at step 0. “ns” is Whisper’s no-speech token and EOT is end-of-transcript. Δ is the trained EOT-row update; $\cos(\Delta, w_{\text{ns}})$ compares it with the native no-speech row, and random p_{95} is the 95th percentile of matched random-direction cosines. The final column tests grafting the no-speech row direction into EOT. Cross-architecture readouts are in supplementary Table 10.

Belrose et al. 2023) applies the model’s frozen final Layer-Norm and unembedding at each depth. Unlike the trained probe, this native abstain direction is non-monotone through the stack (Figure 2a). At the final readout, the message-free margin exceeds the clean-input margin in every evaluated model but remains below the abstention boundary (Figure 2b). The native coordinate therefore orders these evaluated conditions without selecting abstention. Full cross-architecture traces, including Marian’s distinct late-layer trajectory, appear in supplementary Section H.

Native Null-Token Readout

Linear probe performance does not determine the model’s argmax, so we also measure its native output scores. The $\langle \text{nospeech} \rangle$ and EOT logits distinguish non-speech from speech across Whisper scales (Table 1). The same step-0 ordering appears on named developmental subsets for the evaluated non-Whisper AED and translation models: the abstain margin $\text{logit}(\text{EOT}) - \max_t \text{logit}(\text{content}_t)$ distinguishes the evaluated conditions (supplementary Table 10). Distil-Whisper provides an additional readout check in supplementary Table 13.

On message-free inputs, the null-token score often remains below a content token. This gap motivates testing the output coordinate as a calibration target (Section 5); it does not establish why the stock checkpoint learned that score.

Native and External Abstention Scores

The trained update Δ is weakly aligned with the $\langle \text{nospeech} \rangle$ direction. Its cosine is comparable to the upper tail of random directions on every model (Table 1). Grafting the no-speech direction into the EOT row, $w_{\text{EOT}} \leftarrow w_{\text{EOT}} + \lambda(w_{\text{ns}} - w_{\text{EOT}})$, does not suppress normalized lexical output.

Whisper defines `no_speech_prob` at the SOT position, where the no-speech token competes with language tokens. Read there by one shared implementation, it leaves 4.3% residual LOR on the developmental battery under the illustrative 10% speech-side cap, versus 89.8% for thresholding the native EOT margin. Thus the separate native detector is

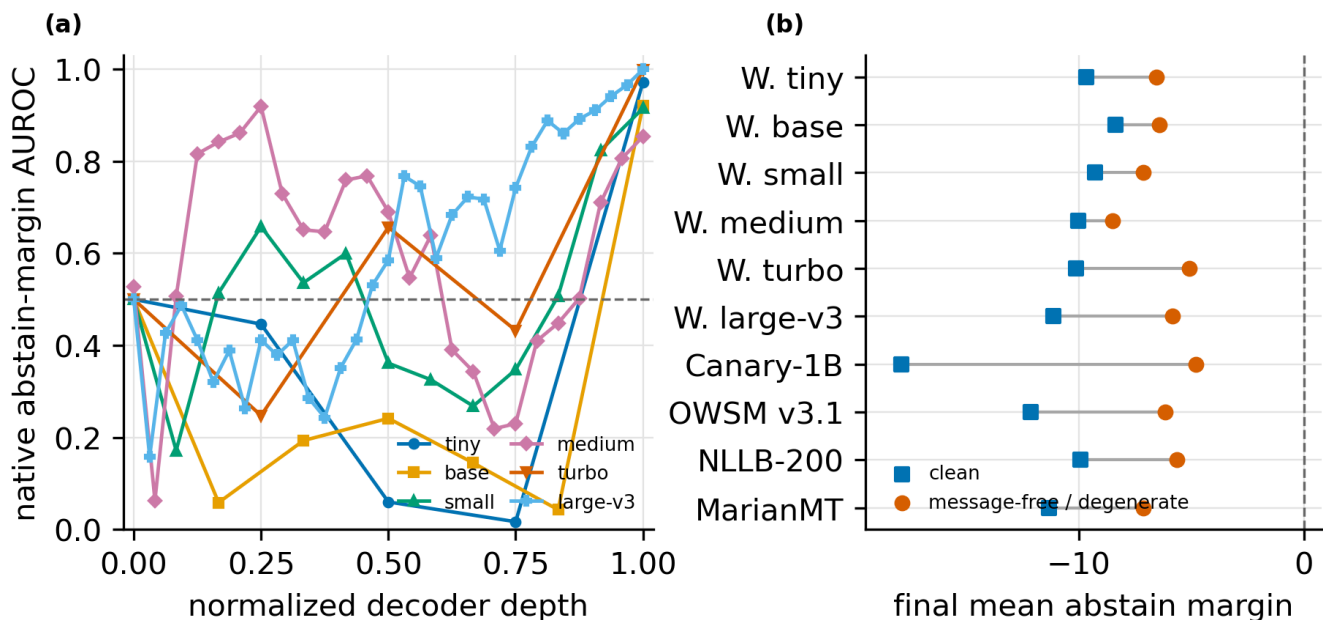


Figure 2: Developmental native null-token readout at decode step 0. (a) Whisper abstain-margin AUROC across the 1060-clip battery and decoder depth. (b) Class-conditional mean final margins on each model’s named developmental subsets; connecting lines join the two means and zero is the abstention boundary. Subset denominators are in the supplement.

Developmental method	LOR	speech rejected/empty
native EOT margin	89.8	≤ 10.0
native no_speech_prob	4.3	≤ 10.0
frozen-state logistic gate	0.0	≤ 10.0
FireRedVAD front-end	1.3	≤ 10.0
stock checkpoint	96.2	0.0
EOT bias $b = 5$	24.3	7.4
trained EOT row	0.2	67.7
analytic graft $\lambda = 2$	0.0	35.6

Table 2: Matched but inspected Whisper-small comparison on 630 non-speech and 430 speech clips (percent). For the four thresholded detectors, the review bundle preserves only that the selected point satisfies the $\leq 10\%$ cap, not the exact speech-side rate. The cap is illustrative and rows are descriptive points, not efficacy or superiority estimates.

useful; the observed failure is specific to the EOT operating point, not an absence of all native abstention signal.

On an identical Whisper-small common-cost set, a frozen-state logistic gate reaches 0.0% LOR and FireRedVAD (FireRed Team 2026) reaches 1.3%, both within the cap. A constant EOT bias is also within the cap (24.3% LOR; 7.4% real-speech empty output), whereas the trained row (0.2%; 67.7%) and analytic graft (0.0%; 35.6%) operate beyond it. The row edits package a state-dependent score change in one checkpoint row, but that is an implementation format rather than a risk-frontier advantage. All arms and denominators appear in supplementary Table 6.

Parameter-Locus Controls

Tuning LayerNorm can suppress message-free output, but a matched-capacity random subnetwork can do so as well across the evaluated learning-rate and seed grid (supplementary Table 17). Their speech costs differ and are sensitive to optimization, especially for Whisper medium. The control therefore supports reachability from several parameter subsets, not a privileged locus or equivalent safety.

Sequence-Level Behavior

Configuration suppression. Removing the stock rule that suppresses EOT at the first output position does not change stock LOR. The rule can block a calibrated EOT decision, however, so all stock and patched comparisons use the same configuration. Details and the diagnostic control are in supplementary Section B.

Position-wise margins. To see where along the sequence the failure lives, we teacher-force each stock model on its own saved hallucinated transcript over non-speech audio and read the abstain margin per position (supplementary Table 19). Stock margins are negative at most measured positions. The calibrated readout increases the margin most at early positions, but turbo and large-v3 remain negative later in the sequence. This result limits the intervention to short, whole-segment abstention. The separate long-form test reduces gap words but increases WER. Correctly cutting speech regions with VAD, rather than merely muting gaps, dominates the row edit on the evaluated long-form set (supplementary Table 21).

5 RQ2: Null-Coordinate Diagnostic Audit

We treat prior shift as a candidate explanation for the step-0 null-token margin, derive its conditional scalar prediction, and test where that explanation fails. We then compare the scalar intervention with two supervised row-based recalibrators.

Conditional Prior-Shift Model

Under label shift, a scalar correction is exact only if the neural margin is calibrated unit-scale posterior log odds. With fixed class-conditionals, that assumption gives $m_{\text{test}} = m_{\text{train}} + \text{logit}(\pi_{\text{test}}) - \text{logit}(\pi_{\text{train}})$ (Saerens, Latinne, and Decaestecker 2002; Lipton, Wang, and Smola 2018). An arbitrary EOT-versus-content margin need not satisfy it; otherwise the scalar is merely a one-coordinate recalibrator. The full proposition, assumptions, and derivation are in supplementary Section A.

Unit base-rate slope and a margin-predicted error-balance bias are therefore applicability checks, not established properties of stock decoders. They neither identify the unknown training prior nor show that prior shift caused stock behavior.

Operating-Point Prediction

A bias β on the null-token logit makes the model abstain at step 0 exactly where the abstain margin exceeds $-\beta$, so the *whole* dose-response curve is predictable from the logged step-0 margin distribution before another decode. We define β^* as the evaluated grid point that minimizes the absolute difference between missed abstentions and false abstentions. It corresponds to the shifted Bayes threshold only under the additional symmetric-margin and equal-cost assumptions.

The margin prediction agrees with Canary at grid resolution and is adjacent to the OWSM empirical point, but it fails on both translation models (supplementary Table 10 and Figure 3). Thus the scalar null-coordinate intervention is effective on these evaluated systems, but the no-fit operating-point predictor is not general. The same fixed subset is used to choose and report each bias, so these dose-response results are descriptive rather than locked-test estimates.

Base-Rate Dependence of Learned Margins

We train each Whisper EOT row at seven non-speech fractions $p \in [0.1, 0.9]$ and read the converged margin (supplementary Table 11). Every per-scale slope is positive and all unadjusted per-scale intervals exclude zero. Some scales include unit slope while others are steeper. A random-effects combination gives slope 1.29 with 95% CI $[0.94, 1.65]$, with substantial heterogeneity ($I^2 = 84\%$). Equivalence to unit slope under the pre-specified $[0.8, 1.2]$ band does not hold, and the random-effects scale trend is not established ($p = 0.118$). Seed replication comprises 18 independently resampled runs and 126 row trainings. These developmental training runs are consistent with positive log-odds dependence, but do not establish a universal unit coefficient, the stock checkpoints' training priors, or out-of-source generalization. The earlier five-point small sweep is retained in supplementary Table 18 for the row-norm diagnostic.

Null-Token Interventions

The trained EOT row is a supervised update to one vocabulary coordinate, not one scalar. It changes hundreds of row parameters. At the common step-50 checkpoint it suppresses message-free output across the full battery (supplementary Table 7); confidence intervals and paired tests are in supplementary Tables 15–16. Checkpoint trajectories and first-zero comparisons are adaptive development analyses, not independent test estimates.

The analytic row graft is also supervised and state-dependent. A logistic probe supplies a direction w , which is added to the EOT row with scale λ . Non-speech calibration clips impose lower bounds on λ , while speech clips impose upper bounds. Their intersection defines an admissible interval, and the reported scale is an interior point of that interval. Supplementary Section C gives pseudocode for the complete procedure.

The analytic calibration pools are disjoint from the reported stress battery and LibriSpeech test-clean. Its clean-speech constraints do not, however, certify a new acoustic distribution. The method performs well on five Whisper scales and regresses on medium (supplementary Table 7). The analytic and trained row directions are weakly aligned, so they should be viewed as different recalibrators of the same output coordinate.

Additional controls compare random output coordinates, EOT suppression, loss weighting, and a full-vocabulary affine calibrator. They support the choice of the null-token coordinate on the evaluated battery; full results are provided in the supplement.

On the inspected battery, the common step-50 row reaches zero LOR on all six Whisper scales; held-out probe WER is unchanged on four scales and rises on turbo and large-v3. The disjointly fitted analytic graft leaves 0.3–8.4% LOR and has sharply model-dependent test-clean cost, including $2.88 \rightarrow 7.60$ WER on medium. The complete developmental table, including admissible intervals and denominators, is supplementary Table 7.

Locked Fabrication–Deletion Sensitivity

Table 3 evaluates the frozen Whisper-small methods on the post-freeze, source-disjoint 300/300 set. Among these five checkpoint conditions, scalar bias $b = 5$ has the lowest cost point estimate on this sample at $\kappa = 0.5, 1$; stock has the lowest point estimate for $\kappa \geq 2$. The trained row never minimizes the displayed grid. It nevertheless moves the measured endpoint sharply. LOR falls from 91.7% to 3.3%, but the intervention incurs 33.05 deleted reference words/min versus stock's 1.58. Its false silence is 8.0% on clean speech, 42.7% at low SNR, and 38.7% on accented speech. These group rates characterize only the trained row and do not support a comparative group-safety claim. The observed ranking reversal shows sensitivity to the displayed weights; sampling uncertainty for cost gaps and crossover location was not quantified. External gates were not decoded on this locked set, so this is neither a superiority test nor a ranking of the best available rejectors.

Together, these results show that editing the reserved output score is a compact way to move the short-segment ab-

Method	F	D	$C_{0.5}$	C_1	C_2	C_5	C_{10}
stock	32.5	1.6	33.3	34.1	35.7	40.4	48.3
trained EOT row	4.3	33.0	20.8	37.3	70.4	169.5	334.8
EOT bias $b = 5$	10.1	17.6	18.9	27.7	45.3	98.0	185.7
EOT bias $b = 6$	6.4	34.6	23.7	41.0	75.8	179.9	353.6
analytic graft $\lambda = 2$	35.4	17.4	44.1	52.8	70.1	122.1	208.8

Table 3: Locked conditional-rate sensitivity $C_\kappa = F + \kappa D$ (lower is better). F is total words retained by the specified lexical normalizer divided by total non-speech audio minutes; D is total reference-word deletions divided by total speech audio minutes. Component values are rounded to one decimal; displayed costs come from unrounded values. Methods and weights were author-frozen before this 300/300 decode. Boldface marks the descriptive minimum on this sample, not statistical superiority or deployment utility; rank uncertainty is unquantified.

stention operating point. They do not establish a privileged causal locus, message-free-specific use of the trained direction, improved risk-coverage over an external detector, or a known training prior for stock checkpoints.

6 RQ3: Boundary Tests Across Evaluated Systems

These developmental analyses test the boundary of the Whisper result; they are not locked replications. We do not assume one mechanism across model families. AED and translation models use EOT or EOS; CTC and RNN-T systems use alignment blanks with different decoding behavior. Multilingual results require multilingual or per-language calibration.

Other AED and Translation Models

Canary-1B and OWSM v3.1 are AED recognizers unrelated to Whisper; NLLB-200 (NLLB Team et al. 2022) and MarianMT (Junczys-Dowmunt et al. 2018; Tiedemann and Thottingal 2020) are encoder-decoder translators from different families. In each model, the step-0 null-token margin distinguishes message-free from meaningful input, and a scalar null-token bias reduces output on the evaluated dose-response subset (supplementary Table 10).

The cost differs substantially. Canary has a low-cost measured point, whereas OWSM truncates real speech as the bias increases. For OWSM, $\beta = 9$ is the empirical error-balance point and $\beta = 10$ is the lower-LOR endpoint. Canary changes LOR $79.3 \rightarrow 0.0\%$ with WER $7.02 \rightarrow 6.94$ at its measured point, whereas OWSM changes $90.7 \rightarrow 1.3\%$ with WER $8.19 \rightarrow 20.9$. The full grid is in supplementary Table 28; these are different operating points, not conflicting estimates.

The translation models show a controllable EOS coordinate but not the same operating-point prediction. Their message-free sources are empty strings, whitespace, punctuation-only strings (e.g., “...” or “!!!”), and random-character strings. Increasing EOS bias reduces retained lexical output on those sources while also reducing real-source length and translation quality: at the reported points, NLLB lexical-output rate changes $27.5 \rightarrow 1.0\%$ while COMET changes $83.4 \rightarrow 80.9$, and Marian’s rate changes $7.0 \rightarrow 0.0\%$ while COMET changes $79.9 \rightarrow 75.6$. Supplementary Tables 10 and 29 report the selected dose responses, BLEU, chrF, and COMET (Rei et al. 2020). SeamlessM4T

and Parakeet are stock checks only and are excluded from the intervention claim.

CTC and RNN-T Systems

CTC and RNN-T blanks are natively unsuppressed, and their non-speech outputs are usually short fragments rather than fluent text. Anchor fine-tuning helps the two strongest CTC systems in the battery, while a blank-bias scalar helps several others. Crossed interventions show that no single method is uniformly safest (supplementary Table 25). These results concern blank-output control and do not establish the AED mechanism for CTC or RNN-T.

Multilingual Evaluation

An English-trained row does not transfer reliably. A separate multilingual row is calibrated on seven language tags and evaluated on eight held-out tags, with Korean as the largest residual. The other model families use one scalar per evaluated language. These are multilingual or per-language calibrations, not zero-shot transfer. On held-out speech, large and turbo retain median empty-output rates of 0.0% and median CER changes of +1.7 and +0.1 points while reducing mean LOR $81.8 \rightarrow 8.3\%$ and $77.6 \rightarrow 5.8\%$. Smaller scales reach low residual LOR at much more abstaining points (median empty output 29.1–84.2%); Korean is the worst residual language at every scale. Full lexical-output and speech-cost rows are in supplementary Tables 23 and 9.

Representation-Editing Baselines

On the inspected Whisper-small training matrix, the 768-parameter EOT row is the smallest update, reaching zero developmental LOR at the first checkpoint without changing probe WER (Table 4). Larger and randomly located updates also reach zero, establishing parameter efficiency rather than safety or a unique locus. AudioSAE and reduced LLT use different procedures and provide no safer common operating point; their full results and cross-architecture Calm-Whisper checks remain in supplementary Tables 26 and 27.

7 Limitations and Scope

We measure and perturb normalized lexical output on whole-segment empty targets, where every retained lexical token

Method	Locus	Params	0% HR at	WER
eot_row	EOT row	+768	50	5.11
LoRA ($r=8$)	q/v proj.	884,736	50	5.5
decoder-only FT	decoder	154M	50	5.6
full FT	all	242M	50	5.7
random-matched	random scal.	95,232	300	5.2
LayerNorm (all)	LN	95,232	150	7.6
Calm-Whisper	3 heads	+590,208	50 ⁺	5.78
AudioSAE $\alpha = 1$	enc. SAE	+113M	no ^a	3.74
AudioSAE $\alpha = 8$	enc. SAE	+113M	$\alpha = 8^a$	100.0
LLT proxy	layer-mix	+12	no ^b	14.2

Table 4: Cost comparison on Whisper small. Training rows report the first zero-HR step and held-out probe WER (stock 5.11); 50⁺ also requires per-model head screening. ^aThe released AudioSAE is evaluated directly on our 630-clip battery and test-clean protocol (stock WER 3.73): $\alpha = 1$ leaves HR at 94.6%, whereas the first zero-HR point, $\alpha = 8$, gives WER 100% (full sweep in supplementary Table 26). ^bThe reduced LLT layer-mixing proxy raises HR 92 $\rightarrow$ 100% and is not a full teacher-distillation reproduction.

is fabricated by construction. Interleaved fabrication is excluded: it requires per-token decisions on non-empty targets, has no label-free detector, and global abstention can harm it.

Lexical-output suppression and false silence require joint evaluation. Our common-cost comparison reports whether an operating point empties at most 10% of real-speech clips, an illustrative cap rather than application utility. Within it, external state and VAD gates achieve lower developmental LOR (Table 2). A row packages a state-dependent score change in one checkpoint coordinate, a format rather than utility advantage. The locked C_κ family reverses the point-estimate ranking under the displayed weights; cost-gap uncertainty is unquantified.

The model-dependent safety ledger (supplementary Tables 20–22) shows tiny and base improving on degraded speech through insertion and substitution reductions despite more deletions, whereas four stronger Whisper scales increase WER and spontaneous-speech costs vary sharply. Medium and OWSM retain narrow low-cost regions. In long-form decoding, the row reduces gap words but raises WER; correctly implemented VAD segmentation dominates both axes. Thus evidence for checkpoint editing is restricted to short, empty-target segments and does not establish superiority over a standard front-end on locked data.

Analytic calibration and evaluation pools are disjoint, but clean-speech constraints do not certify new acoustics. Deployment needs target-domain calibration, a non-speech base-rate estimate, and explicit false-silence cost. Inspected developmental data include stress battery and reduced dose-response subsets. The source-disjoint, collision-audited locked 300/300 cost evaluation includes previously reported test-clean, lacks the intended offline spontaneous stratum, and is not a complete deployment simulation. Missing anonymous artifact locator and scale-specific batch-size/learning-rate settings leave the author-reported freeze and exact runs independently unauditable from PDFs.

No human transcript adjudication supports the headline endpoint. For every evaluated message-free span, an empty lexical reference makes any output remaining after documented non-lexical event-tag removal fabricated by construction, consistent with established non-speech protocols (Barański et al. 2025; Wang et al. 2025). Speech-containing inputs, where hallucination and mistranscription require adjudication, remain excluded.

Current normalization leaves 13 of 15 models invariant, moving Whisper medium 0.31 and OWSM 6.67 points by hiding tag-only, number-only, and wrong-script outputs. The incomplete blinded protocol provides no positive evidence. False silence, the primary foreseeable harm, peaks in locked low-SNR and accented groups. LLT remains a reduced proxy; beam search untested. Avoid small-model use on degraded or spontaneous speech, long-form audio, or when false silence is costlier than fabrication.

8 Conclusion

Developmentally, decoder states and native null-token scores separate the evaluated conditions, but clip-stratified probes do not establish a source-independent message-absence representation. Scalar bias, analytic row graft, and trained-row edits are parameter-compact diagnostic instruments for moving the short-segment operating point, without establishing a privileged mechanism or safer risk frontier.

On the only post-freeze sample, Whisper-small’s trained row moves LOR 91.7% $\rightarrow$ 3.3% and deleted reference words/min 1.58 $\rightarrow$ 33.05. It never minimizes the displayed C_κ grid: bias $b = 5$ has the lowest point estimate for $\kappa = 0.5, 1$, and stock for $\kappa \geq 2$. This descriptive five-condition ranking has unquantified rank uncertainty and no locked comparison with stronger developmental gates.

Results are consistent with abstention miscalibration under some evaluated conditions. Learned margins’ positive, heterogeneous training-log-odds slopes do not establish the classical unit coefficient; failed translation operating-point predictions limit prior-shift explanations. CTC/RNN-T, translation, and multilingual results require family-specific interpretations.

Developmental cross-family, translation, and multilingual boundaries carry model-dependent safety costs; VAD is the evaluated long-form preference. Null-coordinate editing diagnoses fabrication–deletion tradeoff, not safe abstention. Interleaved hallucination, beam search, deployment utility, and zero-shot transfer remain excluded.

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A Extended Validation and Audit Trail

This section reports the post-freeze validation summarized in the main paper. It adds a source-group manifest, a data-role ledger, an untouched source evaluation, one shared cost definition, and seed replication. The values below were generated from per-example result files rather than transcribed into the analysis scripts.

Data Roles and Source Disjointness

The role ledger contains 12 pools assigned to four roles: fit/calibration, selection, developmental reporting, and locked reporting. The developmental battery contains 1060 clips derived from 869 source recordings. Resolving ESC-50 and UrbanSound8K identifiers back to source recordings reveals 21 UrbanSound8K clip pairs with overlapping spans; these pairs stay within a single role. The locked set contains 300 non-speech and 300 speech clips. Disjointness is checked by sample SHA-1, source key, and speaker. As a falsification test, inserting a calibration clip makes the checker refuse to run.

Locked non-speech comprises silence from an unused random seed plus MUSAN, ESC-50, UrbanSound8K, and music recordings absent from all fit and selection pools. Locked speech comprises LibriSpeech test-clean, WHAM!-degraded and low-SNR mixtures, and accented FLEURS. The planned spontaneous stratum was unavailable offline and is marked absent. Test-clean had appeared earlier as a reported benchmark, so it is disjoint from fitting and selection but not previously unseen.

Result	Fit/calibration pool	Selection rule or pool	Reporting pool	Inspected?
Frozen-state probe	Full battery within five clip folds	Cross-validated probe; PCA fitted within each analysis	The same 630 non-speech and 430 speech clips	yes
Trained EOT row	MUSAN calibration half, synthetic silence, LibriSpeech dev-clean	Common step 50; first-zero rows summarize trajectories	Full 630-clip non-speech battery and named speech subsets	yes
Analytic row graft	Disjoint MUSAN half and LibriSpeech dev-clean speakers	Admissible interval on the calibration pool	Stress battery and test-clean, disjoint from fitting	yes
AED/NMT scalar bias	No parameter fit; margins logged on fixed dose-response data	Bias chosen on 150 non-speech and 60 speech clips	The same dose-response subset; MT quality on WMT19	yes
Common-cost comparison	Each method’s stated calibration	Lowest LOR satisfying the illustrative 10% speech-side cap	The same 630 non-speech and 430 speech clips	yes
Multilingual row/scalars	Seven Whisper tags or named language calibration splits	Frozen row/scalars after calibration	Eight held-out tags or held-out language clips	yes
Long-form analysis	Previously calibrated row or fixed VAD configuration	No selection on the reporting clips	40 clips with speech and message-free gaps	yes
Locked C_{κ}	All methods and costs frozen before decoding	No selection on locked outputs	Source-disjoint 300 non-speech / 300 speech clips	no

Table 5: Fit, selection, and reporting roles. “Inspected” means that reporting-pool outputs were available during method development. The locked cost ledger entry is the only post-freeze reporting pool; exact source IDs and computed collisions are author-reported as stored in project manifests that are not linked from the review PDF.

Matched Developmental Cap Comparison

All arms in Table 6 use the same 630 non-speech and 430 speech clips. We mark whether a method is within the illustrative cap of rejecting or emptying no more than 10% of real-speech clips. The first four rows are separately thresholded detectors. Their exact achieved speech-side rates are absent from the available summary, which preserves only that the selected thresholds are within the cap. The last four rows are decoder/checkpoint conditions.

Method	LOR	speech-side rate	within cap
native EOT margin	89.8	≤ 10.0	yes
native no_speech_prob (SOT)	4.3	≤ 10.0	yes
logistic gate (frozen state)	0.0	≤ 10.0	yes
FireRedVAD front-end	1.3	≤ 10.0	yes
stock	96.2	0.0	yes
constant EOT bias $b = 5$	24.3	7.4	yes
trained eot_row	0.2	67.7	no
analytic graft $\lambda = 2$	0.0	35.6	no

Table 6: Matched, inspected Whisper-small comparison (percent). LOR is the normalized lexical-output rate. Speech-side rate is rejection for detectors and empty output for checkpoint interventions. The external detectors have lower developmental LOR while satisfying the coarse cap; this is not a locked efficacy or superiority comparison.

Model	trained EOT row			analytic graft				
	stock LOR	patched LOR	patched WER (stock)	λ^*	$[\lambda_{lo}, \lambda_{hi}]$	in?	LOR	test-clean WER
tiny	94.9	0.0	10.61 (10.61)	1.017	[0.45, 2.28]	yes	0.3	7.54 $\rightarrow$ 8.22
base	99.7	0.0	7.74 (7.74)	0.507	[0.51, ∞)	yes	2.5	5.04 $\rightarrow$ 4.84
small	96.2	0.0	5.11 (5.11)	2.406	[0.53, 10.91]	yes	0.5	3.44 $\rightarrow$ 4.16
medium	86.8	0.0	3.59 (3.59)	0.497	[0.50, ∞)	yes	1.3	2.88 $\rightarrow$ 7.60
turbo	74.3	0.0	3.59 (2.69)	0.661	[0.66, ∞)	yes	3.5	2.16 $\rightarrow$ 2.26
large-v3	77.0	0.0	2.47 (2.41)	0.824	[0.82, ∞)	yes	8.4	2.02 $\rightarrow$ 2.08

Table 7: Developmental trained-row and analytic-graft summary by Whisper scale. Trained-row LOR is on the inspected battery at common step 50; WER is on the held-out dev-clean probe and parentheses give stock WER. The analytic graft uses disjoint calibration pools and reports battery LOR plus test-clean WER. Values and interval bounds are rounded independently.

Locked Fabrication–Deletion Sensitivity

The primary locked endpoint is

$$C_\kappa = F + \kappa D,$$

where F is total words retained by the specified lexical normalizer divided by total non-speech audio minutes and D is total reference-word deletions divided by total speech audio minutes. The effective weight κ combines relative exposure/prevalence and per-word harm; this is a conditional-rate sensitivity index, not expected deployment utility or total WER. Methods, configurations, and $\kappa \in \{0.5, 1, 2, 5, 10\}$ were pre-specified and author-frozen before the 300/300 decode.

Method	F	D	$C_{0.5}$	C_1	C_2	C_5	C_{10}
stock	32.5	1.6	33.3	34.1	35.7	40.4	48.3
trained EOT row	4.3	33.0	20.8	37.3	70.4	169.5	334.8
constant EOT bias $b = 5$	10.1	17.6	18.9	27.7	45.3	98.0	185.7
constant EOT bias $b = 6$	6.4	34.6	23.7	41.0	75.8	179.9	353.6
analytic graft $\lambda = 2$	35.4	17.4	44.1	52.8	70.1	122.1	208.8

Table 8: Locked C_κ (lower is better). F and D are rounded to one decimal; costs use unrounded components. Boldface marks the descriptive minimum on this 600-clip sample. Rank uncertainty is unquantified, the weights are illustrative, and external gates were not run on this set.

The trained row reduces clip-level LOR from 91.7% to 3.3% (zero on silence, music, events, and urban audio), but deletes 33.05 reference words/min versus stock’s 1.58. Its groupwise false-silence rate is 8.0% on clean speech, 42.7% on low-SNR speech, and 38.7% on accented speech. These worst groups are reported rather than averaged away. They characterize the trained row only; groupwise rates for every locked condition are not available in the displayed result record.

Held-Out Multilingual Speech Cost

For each language, 40 clips calibrate the multilingual row and 60 disjoint clips report hallucination, real-speech CER, and empty output.

Model	stock LOR	fixed LOR	worst-language LOR	median empty	median Δ CER
large-v3	81.8	8.3	pt: 20.0	0.0	1.7
turbo	77.6	5.8	es: 11.7	0.0	0.1
base	92.3	3.1	ko: 46.7	29.1	24.7
small	79.0	2.0	ko: 8.3	47.5	46.4
tiny	71.8	3.8	ko: 56.7	72.5	56.7
medium	75.0	0.3	ko: 3.3	84.2	84.2

Table 9: Held-out 15-language Whisper evaluation (percent). Large-v3 and turbo retain real speech at the calibrated point; smaller scales reach low mean LOR by abstaining heavily. Korean is the worst residual language at all six scales.

Model	Task	AUROC ₀	lexical LOR	Speech/MT change	β	β^* pred.	β^* emp.
Canary-1B	ASR	1.000	79.3 $\rightarrow$ 0.0	WER 7.02 $\rightarrow$ 6.94	8	8.0	8.0
OWSM v3.1	ASR	0.998	90.7 $\rightarrow$ 1.3	WER 8.19 $\rightarrow$ 20.9	10	10.0	9.0
NLLB-200	NMT	0.999	27.5 $\rightarrow$ 1.0	COMET 83.4 $\rightarrow$ 80.9	6	3.5	6.0
MarianMT	NMT	0.911	7.0 $\rightarrow$ 0.0	COMET 79.9 $\rightarrow$ 75.6	6	9.5	3.0

Table 10: Developmental scalar null-token-bias summary (percent except β). AED dose response uses 150 non-speech and 60 speech clips; NMT quality uses real-source WMT19. β^* is selected and reported on the same grid, so these are descriptive rather than locked estimates. COMET is multiplied by 100.

Conditional Prior-Shift Derivation

Let $z = 1$ mark message-free input (abstain), $z = 0$ input carrying a message (emit), and $\pi = P(z=1)$. The decoder’s step-0 score is the abstain margin $m(x)$. Changes in acoustics, language tags, or decoder context violate the fixed-class-conditional premise below.

Proposition 1 (Calibrated-margin prior correction) *Assume label shift with fixed $p(x \mid z)$ and a unit-scale calibrated training margin,*

$$m_{\text{train}}(x) = \log \frac{p(x \mid z = 1)}{p(x \mid z = 0)} + \text{logit}(\pi_{\text{train}}).$$

Then $m_{\text{test}}(x) = m_{\text{train}}(x) + \text{logit}(\pi_{\text{test}}) - \text{logit}(\pi_{\text{train}})$; for our margin, the constant can be added to the EOT logit.

This is the classical Bayes identity $\text{logit } P(z=1 \mid x) = \log \text{LR}(x) + \text{logit}(\pi)$ (Saerens, Latinne, and Decaestecker 2002; Elkan 2001; Lipton, Wang, and Smola 2018). It is not a theorem about an arbitrary neural EOT-versus-content margin. If the neural margin approximates the assumed score, three diagnostics follow: a row trained at non-speech fraction p should install a margin linear in $\text{logit}(p)$ with unit slope; a deployment scalar is the classical prior correction; and, under an additional one-step symmetric-margin and equal-cost approximation, an error-balance bias can be predicted from step-0 margins. The first and third are applicability checks, not evidence that prior shift caused the stock behavior.

Model	slope	95% CI	r
tiny	0.96	[0.41, 1.50]	0.90
base	0.80	[0.52, 1.08]	0.96
small	1.05	[0.65, 1.45]	0.95
medium	1.53	[1.01, 2.06]	0.96
turbo	1.89	[1.25, 2.52]	0.96
large-v3	1.69	[1.18, 2.21]	0.97
pooled RE	1.29	[0.94, 1.65]	n.a.

Table 11: Developmental replicated analysis of learned abstention margin versus $\text{logit}(p)$ over seven training fractions per scale. These run-level intervals summarize the evaluated trainings; heterogeneity is high and equivalence to unit slope is not established.

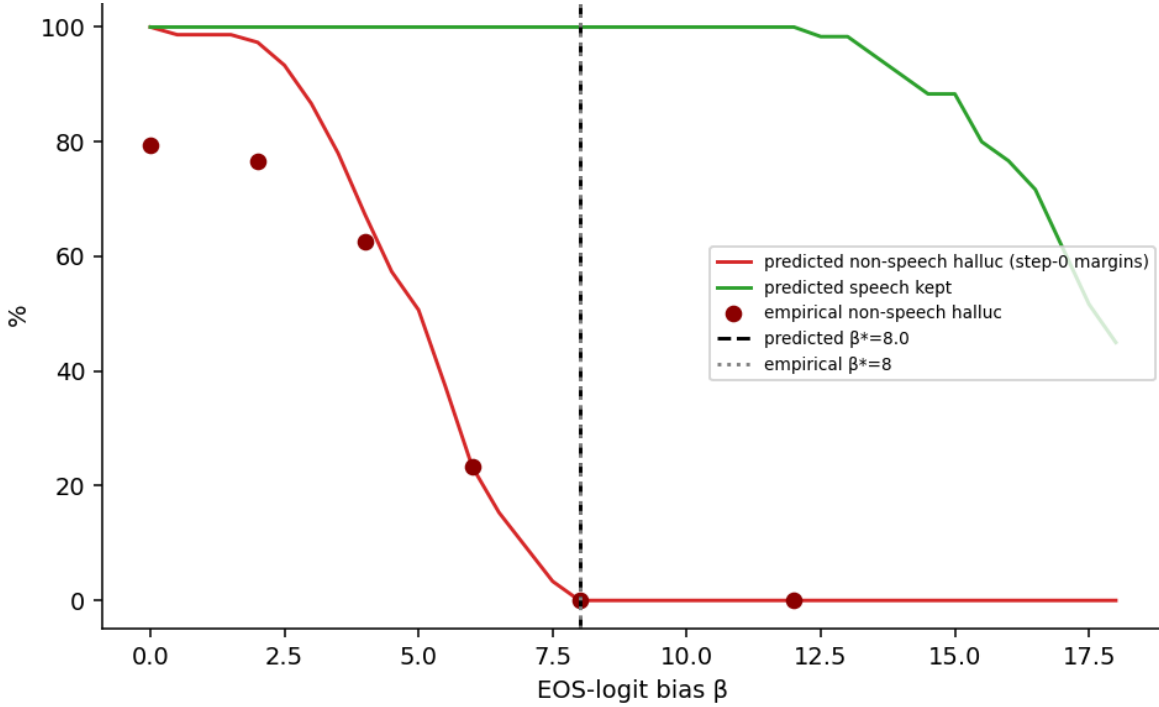


Figure 3: Developmental Canary-1B dose response on 150 non-speech and 60 speech clips. Curves are predicted from step-0 margins; points are measured decodes. The dashed line marks the descriptive error-balance grid point, where missed and false abstentions are closest.

Statistical, Scoring, and Protocol Audits

The six base-rate slopes are combined with a random-effects model because heterogeneity is high ($\tau^2 = 0.163$, $I^2 = 84\%$). The pooled slope is 1.29 with 95% CI [0.94, 1.65]; the scale trend is +0.214 per scale step ($t = 1.99$, 4 df, $p = 0.118$). No per-scale departure from unity survives Holm or Benjamini–Hochberg correction, but equivalence to unity under [0.8, 1.2] also fails. Seed replication uses 18 independent resamples with five seeds on small/medium/large and three on tiny/base/turbo, giving 126 row trainings. Batch-size recipe sensitivity is logged separately from seed variation. Separately, the reported Wilson intervals and McNemar tests use clips as units on an inspected battery. They are finite-battery summaries, are not clustered by source recording, and do not quantify corpus-transfer or method-selection uncertainty. The PDF-level record does not establish whether the 21 overlapping-span UrbanSound pairs share cross-validation folds.

Normalization sensitivity is zero on 13 of 15 cached models, 0.31 percentage points on Whisper medium, and 6.67 points on OWSM. For OWSM, the current normalizer hides 34 tag-only, 8 number-only, and 7 wrong-script clips. This is why the main paper treats lexical normalization as a measurement boundary rather than semantic adjudication.

The audit infrastructure includes a blinded 150-clip sheet with opaque arm IDs, shuffled rows, six dimensions for two annotators, and an adjudication column. The sheet is prepared; annotation has not been completed. For multilingual NMT, a point is marked within cap only if at least 90% of genuine-source translations remain non-empty. NLLB Chinese and Marian Spanish have no point within that cap and are marked as failures rather than included in the mean.

B Implementation and Procedural Checks

Common training recipe. Training uses an example-balanced 50/50 anchor mix: MUSAN-training-half noise plus synthetic silence with an empty target, and lower-cased LibriSpeech dev-clean speech with its transcript. Runs use weight decay 0, gradient clipping at 1.0, 400 steps, and evaluation every 50 steps. AdamW is used below 50M trainable parameters and Adafactor above; the EOT row uses learning rate 10^{-3} . Decoder inputs are passed explicitly and all stock/patch evaluations use greedy temperature-0 decoding with `begin_suppress_tokens=None`. The reported step-50 row is fixed across scales; “first zero” is adaptive and descriptive. Scale-specific batch sizes and the broader learning-rate grid remain in the prepared configuration inventory and are not independently recoverable from this PDF alone.

Four HuggingFace-pipeline defaults each silently reproduce “the model resists fine-tuning” without any involvement of the model. We list the symptom, the diagnostic, and the fix.

Prefix masking. Masking the forced prefix in `labels` with `-100` while letting the collator rebuild `decoder_input_ids` via `shift_tokens_right` turns every `-100` into the pad token (EOT), so the decoder trains on the garbage context `[sot, eot, eot, ...]`. Diagnostic: teacher-forced loss near $\ln V \approx 10.7$. Fix: pass `decoder_input_ids` explicitly.

Begin-suppress ban. `begin_suppress_tokens` forbids EOT as the first token, so a model that has perfectly learned $P(\text{EOT} \mid \text{noise}) \approx 1$ still emits content; on tiny, LayerNorm tuning evaluated with the ban enabled pushed hallucination from 89% to 100%. Fix: evaluate fine-tuned models with `begin_suppress_tokens=None` (verified a no-op on stock models).

Token-averaged loss. A non-speech example contributes one target token (EOT) against roughly forty for a speech example, so token averaging gives the anti-hallucination gradient $\sim 2\%$ of the weight; $-\log P(\text{eot})$ barely moves ($7.3 \rightarrow 6.4$ over 400 steps). Fix: example-balanced loss (mean over per-example mean token cross-entropy).

Reentrant checkpointing. With `use_reentrant=True` (the default), parameters inside checkpointed blocks whose inputs do not require grad receive no gradient: on turbo only the two final LayerNorms out of 156 tensors actually trained. Fix: `use_reentrant=False`.

Selection and scoring checks. Checkpoint selection by minimum clean-probe WER is blind to over-abstention and prefers late, destructive checkpoints (up to 62% of spontaneous out-of-domain speech left empty; Table 22); selection must include out-of-domain speech. And scorers must strip non-speech tags (e.g. `[music]`) before counting fabricated letters, or hallucination is over-counted on models that emit tags.

The sixteen regimes. full; decoder-only; encoder-only; LoRA ($r=8$, q/v); final decoder layer; LayerNorm all / γ only / β only / encoder-only / decoder-only / final pair; BitFit (all biases except LayerNorm); `eot_row` (the EOT unembedding row); random-matched (random scalars via gradient mask, exactly the LayerNorm count, LayerNorm excluded); random-colocated (random scalars in the attention/MLP tensors of the same blocks); calm-heads (top-3 decoder self-attention heads by ablation screening).

C Analytic Row-Graft Procedure

The analytic graft is a supervised linear recalibrator of one output coordinate. It is not a scalar prior-odds correction. The following pseudocode records the reported construction.

1. Collect frozen decoder states from labeled calibration data. Positive states are non-speech at the first output position and genuine end-of-utterance positions. Negative states are mid-utterance speech positions.
2. Fit a logistic direction w to those states.
3. For each non-speech clip, convert the requirement that EOT beat every content logit into a lower bound on λ . Set λ_{lo} to the largest lower bound.
4. For each speech clip and mid-utterance position, convert the requirement that EOT remain below the correct content decision into an upper bound. Set λ_{hi} to the smallest upper bound.
5. Reject the graft if $\lambda_{lo} > \lambda_{hi}$. Otherwise choose $\lambda^* = \sqrt{\lambda_{lo}\lambda_{hi}}$ when the upper bound is finite and use λ_{lo} when it is unbounded.
6. Apply $w_{\text{EOT}} \leftarrow w_{\text{EOT}} + \lambda^* w$ and evaluate on the designated reporting pool.

The interval uses only the calibration pool. Because its speech constraints are clean, feasibility does not certify degraded or out-of-domain speech.

D Reporting and Reproducibility Details

The fit, selection, and reporting pools are recorded in Table 5. This section fixes the terminology used across the detailed result tables and lists the contents of the reproducibility archive.

Subset-dependent terminology. `eot_row@50` always denotes the common step-50 trained-row checkpoint evaluated on the full battery. “First zero” denotes the first inspected checkpoint reaching zero hallucination in a training trajectory and is used only as an adaptive descriptive summary. For OWSM, $\beta = 9$ is the empirical error-balance point, while $\beta = 10$ is the lower-hallucination point. Every WER is interpreted only with the denominator and source group named in its table.

Artifact inventory and release status. The author-reported prepared artifact inventory contains immutable model identifiers and revisions, environment lockfiles, source-level split manifests and checksums, normalization and tag-removal code, random seeds, training and decoding configurations, checkpoint and threshold selection logs, learned rows and biases, per-example stock and patched outputs, and one reproduction command per table. It resolves short labels including `wav2vec2-large`, `HuBERT-large`, `WavLM-100h`, `wav2vec2-base`, `Conformer-CTC`, `Emformer-RNN-T`, and `Conformer-RNN-T`. A stable anonymous locator and snapshot hash are not included in the current review bundle, so public availability is not asserted here. The authors intend to release the artifacts upon publication, subject to upstream model and dataset licenses. Consequently, the freeze, exact revisions, scale-specific batch sizes, broader learning-rate grid, and one-command table reconstruction cannot be independently checked from the review PDFs.

Data use and foreseeable harm. The experiments use cited public audio and translation corpora; source audio remains governed by its original license, and the release inventory redistributes only material permitted by those terms. No new human-subject data were collected. The principal foreseeable harm is false silence: the locked row suppresses 42.7% of low-SNR and 38.7% of accented speech in the measured groups. These disparities motivate the explicit non-deployment recommendation for the row at that operating point.

E Deployment Checklist

1. Build a target-domain calibration set containing message-free, clean-speech, and degraded-speech inputs, with no overlap with final evaluation.
2. Estimate the expected message-free base rate only if the scalar prior-shift model is appropriate; otherwise select a point from a labeled cost curve.
3. Specify the relative cost of fabricated output and false silence, then report both at the selected point.
4. Verify that the decoder permits the null token at the intended output position and that stock and patched configurations are identical.
5. Recalibrate after changes in acoustic domain, language tag, decoder policy, segmentation, or long-form context. Do not extrapolate the single-segment results to interleaved hallucination.

F Stress Set, Statistics, and Spontaneous Speech

Category	Source	n
silence_pure	synthetic zero + dither	40
silence_room	pink noise (room tone)	40
noise_stationary	MUSAN noise	150
noise_events	ESC-50	150
noise_urban	UrbanSound8K	150
music	MUSAN music (instrumental)	100
speech_speed	FLEURS, time-stretched	150
overlap_2/3spk	FLEURS speaker mixes	100
speech_noise	FLEURS + MUSAN	120
speech_clean	FLEURS _{en_us}	60
Total		1060

Table 12: The 1060-clip stress battery (3.1 h, 16 kHz mono, clips ≤ 30 s; seed 20260611). Non-speech strata carry an empty reference; speech strata are FLEURS-derived. Exclusions: UrbanSound8K classes that may contain words (children_playing, street_music) and vocal MUSAN tracks are dropped. Speech sources are RMS-normalized to -20 dBFS.

Model	Type	test-clean	Lexical-output rate (%) by stratum						
		WER %	silence	room	noise	events	urban	music	all
Whisper tiny	AED	7.54	100	100	86	96	99	97	94.9
Whisper base	AED	5.04	100	100	100	99	100	100	99.7
Whisper small	AED	3.44	100	50	97	100	100	100	96.2
Whisper medium	AED	2.88	0	0	99	100	100	98	86.8
Whisper turbo	AED	2.16	25	42	82	67	83	94	74.3
Whisper large-v3	AED	2.02	25	25	88	60	97	97	77.0
Canary-1B	AED	1.44	100	100	74	89	97	19	77.5
OWSM v3.1	AED	2.42	78	100	91	92	93	99	92.7
wav2vec2-large	CTC	1.73	0	0	19	23	4	21	14.4
wav2vec2-base	CTC	3.26	0	0	33	37	13	69	30.8
HuBERT-large	CTC	1.97	0	0	75	82	35	100	61.4
WavLM-100h	CTC	6.57	0	0	36	47	20	74	36.2
Conformer-CTC	CTC	2.08	0	0	15	15	7	13	11.1
Emformer-RNN-T	RNN-T	4.43	0	0	21	23	5	83	24.8
Conformer-RNN-T	RNN-T	1.65	0	0	5	3	0	0	1.9

Table 13: Stock behavior of the fifteen-model core zoo. LibriSpeech test-clean WER (Canary-1B and OWSM v3.1 on 500 utterances, others on 2620) validates the pipeline. Lexical-output rate is the share of non-speech clips with at least one retained letter after normalization. Exact repository revisions are author-reported as listed in a prepared model manifest that is not linked from the review PDF.

Use	Evaluation set	Reason
zoo/stats	full 630 NS + 430 speech	headline rates/CIs
deployable ROC	fixed 300 NS + 430 speech	threshold sweeps
AED/NMT bias	fixed 150 NS + 60 speech	dose response
degraded ledger	fixed ID degradations	error alignment
spontaneous cost	Earnings-22/MUSAN-speech	external domain
long form	40 clips, 74 min	chunk boundaries

Table 14: Evaluation-subset map. Subsets differ because large-model threshold sweeps are substantially more expensive than one-pass headline evaluation. Every intervention is compared with stock re-decoded on the same subset; rates are not compared across rows. The first three subsets are fixed and nested only where their listed counts coincide. NS denotes non-speech.

Model	stock		eot_row@50		analytic	
	LOR	95% CI	LOR	95% CI	LOR	95% CI
tiny	94.9	[92.9, 96.4]	0.0	[0.0, 0.6]	0.3	[0.1, 1.2]
base	99.7	[98.8, 99.9]	0.0	[0.0, 0.6]	2.5	[1.6, 4.1]
small	96.2	[94.4, 97.4]	0.0	[0.0, 0.6]	0.5	[0.2, 1.4]
medium	86.8	[84.0, 89.2]	0.0	[0.0, 0.6]	1.3	[0.6, 2.5]
turbo	74.3	[70.7, 77.5]	0.0	[0.0, 0.6]	3.5	[2.3, 5.2]
large-v3	77.0	[73.5, 80.1]	0.0	[0.0, 0.6]	8.4	[6.5, 10.8]

Table 15: Normalized lexical-output rates (LOR, %; $n=630$ non-speech clips) with clip-level Wilson 95% intervals (Wilson 1927). These conditional finite-battery summaries are not source-clustered or corpus-transfer intervals. The trained row is evaluated at common step 50; paired tests are in Table 16.

Model	b	c	exact p	Holm p	OR
tiny	598	0	2×10^{-180}	8×10^{-180}	1197
base	628	0	2×10^{-189}	1×10^{-188}	1257
small	606	0	8×10^{-183}	4×10^{-182}	1213
medium	547	0	4×10^{-165}	1×10^{-164}	1095
turbo	468	0	3×10^{-141}	3×10^{-141}	937
large-v3	485	0	2×10^{-146}	4×10^{-146}	971

Table 16: Exact paired McNemar tests (McNemar 1947) for stock versus eot_row@50 on all 630 non-speech clips. b is stock-nonempty/patch-empty and c the reverse. Holm correction is over the six-scale family; OR uses the Haldane correction because every $c = 0$. All learned-patch LORs are 0.0% (Wilson CI [0.0, 0.6]). The independently decoded analytic patch likewise has $c = 0$ on every scale and exact $p < 10^{-130}$; its per-scale LORs are in Table 15. These clip-level tests are conditional on the inspected battery and do not address method-selection or source-transfer uncertainty.

Model	grid runs	max residual LOR	probe-WER range
tiny	9	0.0	11.1–11.9
base	9	0.0	8.3–9.3
small	9	2.0	5.1–9.0
medium	9	0.0	3.9–63.1
turbo	9	0.0	2.4–11.4
large-v3	9	0.0	2.2–3.7

Table 17: Matched-capacity random-subnetwork verification on all six scales: three learning rates by three mask seeds per scale. Max residual LOR is the worst final normalized lexical-output rate across the nine runs. Reachability is scale-wide (at most 2%), but WER safety is not: medium is acutely learning-rate-sensitive. This supports no privileged parameter locus, not interchangeable operating points.

p (non-speech)	$\text{logit}(p)$	abstain margin	$\ \Delta\ $ row
0.20	−1.39	12.67	0.359
0.35	−0.62	14.98	0.379
0.50	+0.00	14.46	0.373
0.65	+0.62	14.63	0.366
0.80	+1.39	16.10	0.353

Table 18: Raw base-rate sweep behind Table 11 (Whisper small): converged step-0 abstain margin and unembedding row-delta norm at each training non-speech fraction p . Fitted slope 0.98 in $\text{logit}(p)$ ($r = 0.85$; 95% CI [−0.12, 2.09], $n=5$); the row norm stays nearly constant; the row controls the gain while the base rate changes the threshold.

Model	Stock margin			Patched margin		
	0	10	30	0	10	30
tiny	−6.4	−1.3	−3.4	+14.5	+6.5	+4.9
base	−6.2	−1.0	−2.6	+15.2	+6.1	+5.9
small	−7.8	−3.1	−2.5	+16.2	+4.1	+5.8
medium	−8.0	−3.3	−2.5	+18.9	+5.4	+10.2
turbo	−4.6	−3.4	−4.9	+12.0	+2.2	−0.8
large-v3	−4.9	+0.3	−8.0	+9.8	+6.3	−4.8

Table 19: Position-wise abstain margin $\text{logit}(\text{EOT}) - \max_t \text{logit}(\text{content}_t)$ at emitted positions 0, 10, 30, from teacher-forcing each stock model on its own hallucinated transcript. Stock margins are negative almost everywhere; the patch lifts them most at early positions, and residual late negativity on turbo and large-v3 explains why the position-0 intervention does not transfer directly to long-form decoding.

Model	Δ Sub	Δ Del	Δ Ins	Δ WER
tiny	−16.1	+24.3	−17.7	−9.5
base	−7.8	+12.4	−13.7	−9.1
small	−3.8	+10.0	−0.8	+5.5
medium	−3.8	+13.4	−1.3	+8.4
turbo	−2.9	+12.4	−0.7	+8.8
large-v3	−3.0	+15.1	−0.6	+11.6

Table 20: Fixed-subset decomposition of the degraded-speech WER change under the trained EOT row, in percentage points. Tiny and base remove enough substitutions and insertions to offset added deletions; the four stronger scales over-delete.

Variant	gap words/min	deletions	WER
stock	62.0	2.5	7.4
eot_row@50	27.6	14.5	21.1
VAD mute + stock	81.2	2.2	16.0
VAD segmentation + stock	4.7	0.3	4.5
boundary sweep	26.3–72.9	1.1–15.6	22.0–54.6

Table 21: Whisper turbo on 40 clips (38.7 gap-minutes). The earlier 81.2 row muted gaps but decoded fixed windows; it was mislabeled as segmentation. Correct VAD segmentation cuts and independently decodes speech regions, dominating the null patch on both frontier axes. Rates are percent except gap words/min.

Model	LayerNorm @ best-zero step			eot_row @ 50		
	WER	deletions	empty/100	WER	deletions	empty/100
tiny	23.7 → 50.9	5.7 → 41.8	20	23.7 → 63.1	5.7 → 55.6	22
base	16.5 → 21.4	4.2 → 9.9	7	16.5 → 20.6	4.2 → 9.6	2
small	13.2 → 23.4	5.1 → 15.3	10	13.2 → 23.1	5.1 → 16.6	5
medium	11.7 → 12.1	3.8 → 4.2	3	11.7 → 14.9	3.8 → 8.0	0
turbo	13.8 → 25.0	4.0 → 15.0	30	13.8 → 13.4	4.0 → 4.3	27
large-v3	12.3 → 24.5	3.4 → 18.9	62	12.3 → 16.3	3.4 → 9.0	25

Table 22: Spontaneous-speech cost and checkpoint-selection risk. Earnings-22 (Del Rio et al. 2022) WER and deletion rates (stock→patched, %) and empty outputs per 100 spontaneous MUSAN-speech clips, for the LayerNorm regime at its best zero-hallucination checkpoint and eot_row at step 50. Clean-probe checkpoint selection is blind to this: late checkpoints leave up to 62% of spontaneous speech empty (large-v3, LayerNorm at step 350); tiny pays +27→+39 pp WER under any zeroing checkpoint; medium (LayerNorm) and turbo (eot_row) are nearly free.

G Cross-Lingual and Cross-Architecture Detail

This appendix carries the held-out-language detail behind Section 6, the language-identification repair (Table 24), and standard MT quality under EOS bias (Table 29). Unexecuted cross-architecture baseline placeholders have been removed rather than presented as evidence.

Lang split		stock	patched	WER
en	pilot	63	1.3	6.3
ru	pilot	100	0.8	n.a.
de	pilot	81	1.0	13.9
ja	pilot	100	2.0	n.a.
hi	pilot	86	0.2	n.a.
zh	pilot	100	0.7	n.a.
ar	pilot	100	0.5	n.a.
fr	held out	18	1.5	19.7
es	held out	90	1.7	8.5
it	held out	80	1.7	10.8
nl	held out	36	1.8	24.5
pt	held out	90	3.0	10.6
pl	held out	92	2.5	24.1
tr	held out	81	3.0	20.3
ko	held out	92	13.3	n.a.

Table 23: Whisper-small multilingual EOT row: hallucination % on seven calibration tags and eight held-out tags, plus real-speech WER where word-based scoring was available. Seven held-out tags remain at 1.5–3.0%; Korean is the visible 13.3% exception. This is multilingual calibration followed by language holdout, not English zero-shot transfer.

Lang	Script	langid (stock→fix)	WER (stock→fix)
en	Latin	0.38 → 1.00	0.048 → 0.048
de	Latin	0.28 → 0.96	0.116 → 0.116
fr	Latin	0.59 → 0.97	0.170 → 0.170
es	Latin	0.19 → 0.91	0.081 → 0.081
it	Latin	0.06 → 0.80	0.124 → 0.124
nl	Latin	0.56 → 0.95	0.188 → 0.188
pt	Latin	0.47 → 0.91	0.101 → 0.101
pl	Latin	0.32 → 0.97	0.202 → 0.202
tr	Latin	0.07 → 0.90	0.131 → 0.131
ru	Cyrillic	0.32 → 0.89	0.141 → 0.141
ar	Arabic	0.51 → 0.95	0.332 → 0.332
hi	Devanagari	0.22 → 0.96	0.654 → 0.654
ja	Kana	0.06 → 0.97	n.a. [†]
ko	Hangul	0.06 → 0.92	0.225 → 0.225
zh	Han	0.24 → 0.98	n.a. [†]

Table 24: One row per language token restores Whisper small’s short-clip (2 s) language identification without touching transcription: langid accuracy and word WER (stock→fix) across fifteen FLEURS languages and seven scripts. Mean langid rises 0.29 → 0.94; WER is bit-identical (the edit is scoped to the detection step). [†]ja/zh word-WER is ≈ 1 by construction (no word boundaries); only its invariance under the fix matters.

Model	Intervention	Halluc. (%)	WER
wav2vec2 (CTC)	anchor FT	14.4 → 0.8	2.30 → 2.30
HuBERT (CTC)	anchor FT	61.4 → 0.5	2.24 → 4.66
WavLM (CTC)	bias $\beta=2$	36.2 → 14.0	9.15 → 10.77
wav2vec2-base (CTC)	bias $\beta=2$	30.8 → 16.8	4.15 → 5.72
Conformer (CTC)	bias $\beta=2$	11.1 → 4.3	2.75 → 3.82
Emformer (RNN-T)	bias $\beta=4$	24.8 → 7.3	7.13 → 8.14
Conformer (RNN-T)	none needed	1.9	1.65

Table 25: The blank is the CTC/RNN-T null token. Hallucination and WER are stock→patched (%); “bias” is a single blank-logit scalar. Strong CTC systems reach near-zero via anchor fine-tuning; weaker systems reduce monotonically under one scalar.

Released AudioSAE-small condition	HR	95% CI	WER	Δ WER
stock (SAE disabled)	96.2	[94.4, 97.4]	3.73	+0.00
reconstruction ($\alpha = 0$)	96.2	[94.4, 97.4]	3.41	−0.32
steering $\alpha = 1$	94.6	[92.6, 96.1]	3.74	+0.01
steering $\alpha = 2$	95.1	[93.1, 96.5]	61.33	+57.60
steering $\alpha = 4$	48.9	[45.0, 52.8]	188.57	+184.85
steering $\alpha = 8$	0.0	[0.0, 0.6]	100.00	+96.27

Table 26: Common-axis evaluation of the released AudioSAE-Whisper-small final-layer encoder SAE (Aparin et al. 2026b). We select the top 128 features by a logistic regression fit only on training pools and apply the released steering recipe. At the paper-strength setting $\alpha = 1$, HR barely moves; increasing strength destroys transcription before suppressing fabrication. Zero HR is reached only when WER is 100%. One SAE layer replicated across the twelve encoder layers contributes 113.3M parameters, versus 768 for the EOT row.

Model	Method	Screening	Params	HR	WER
Canary	stock	—	—	77.5	1.66
Canary	Calm top-8 head ablation	192 passes	0	72.9	1.70
Canary	EOS scalar, $\beta = 8$	none	1	0.0	6.94
OWSM	stock	—	—	100.0	16.11
OWSM	Calm top-8 head ablation	288 passes	0	100.0	16.08
OWSM	EOS scalar, $\beta = 9$	none	1	4.7	12.79

Table 27: Cross-architecture Calm-Whisper route versus a one-scalar EOS edit. Screening all candidate decoder heads finds a maximum step-0 $\Delta P(\text{EOS})$ of only 0.0006 on Canary and 0.0001 on OWSM; ablating the selected heads consequently has little or no effect. Canary uses 630 non-speech clips and 500 test-clean utterances. The slower OWSM ablation uses 300/150 with $\text{maxlenratio}=0.4$, so its absolute WER is not comparable with other OWSM tables, but its stock/ablation comparison is paired. Scalar rows report their independently measured operating points.

OWSM EOS bias β	HR	WER
0.0	90.7	8.19
2.0	84.0	8.19
4.0	72.7	8.19
6.0	58.0	8.92
8.0	16.7	9.43
8.5	9.3	11.04
9.0	4.7	12.79
9.5	4.0	15.13
10.0	1.3	20.91
11.0	0.0	48.76
12.0	0.0	89.69

Table 28: Fine OWSM dose response on the fixed 150 non-speech plus 60 speech subset. The bias is a monotone hallucination control, but there is no free operating point: the empirical balance point is $\beta = 9$, whereas the 1.3% HR headline at $\beta = 10$ more than doubles WER.

Model	β	BLEU	chrF	COMET	nonempty	length
NLLB en→ru	0	29.0	54.4	83.4	100	91
NLLB en→ru	6	25.2	50.8	80.9	100	82
NLLB en→ru	8	13.9	37.4	70.3	100	54
Marian en→de	0	37.7	63.7	79.9	100	97
Marian en→de	6	34.5	59.0	75.6	100	87
Marian en→de	10	0.0	5.9	32.2	87	8

Table 29: Real-source WMT19 translation quality under EOS bias. BLEU (Papineni et al. 2002), chrF (Popović 2015), and COMET (Rei et al. 2020) (`wmt22-comet-da`, multiplied by 100) decline at the working point ($\beta = 6$) and decline sharply under stronger early-termination bias. Nonempty and length are percentages.

H Cross-Architecture Logit-Lens Detail

Figure 4 expands the final-margin summary in Figure 2b into complete decoder-depth traces for the two non-Whisper speech AEDs and two translation models. Canary, OWSM, and NLLB keep the mean null-token margin below zero throughout. Marian is qualitatively different: EOS is the mean argmax through the middle stack for all three source conditions, but the last decoder block reverses that preference, most strongly on clean sources. The common final failure is therefore compatible with different internal trajectories. Three models do not promote abstention, while Marian demotes it late. We consequently focus on the final readout decision rather than claim one universal layer-wise mechanism.

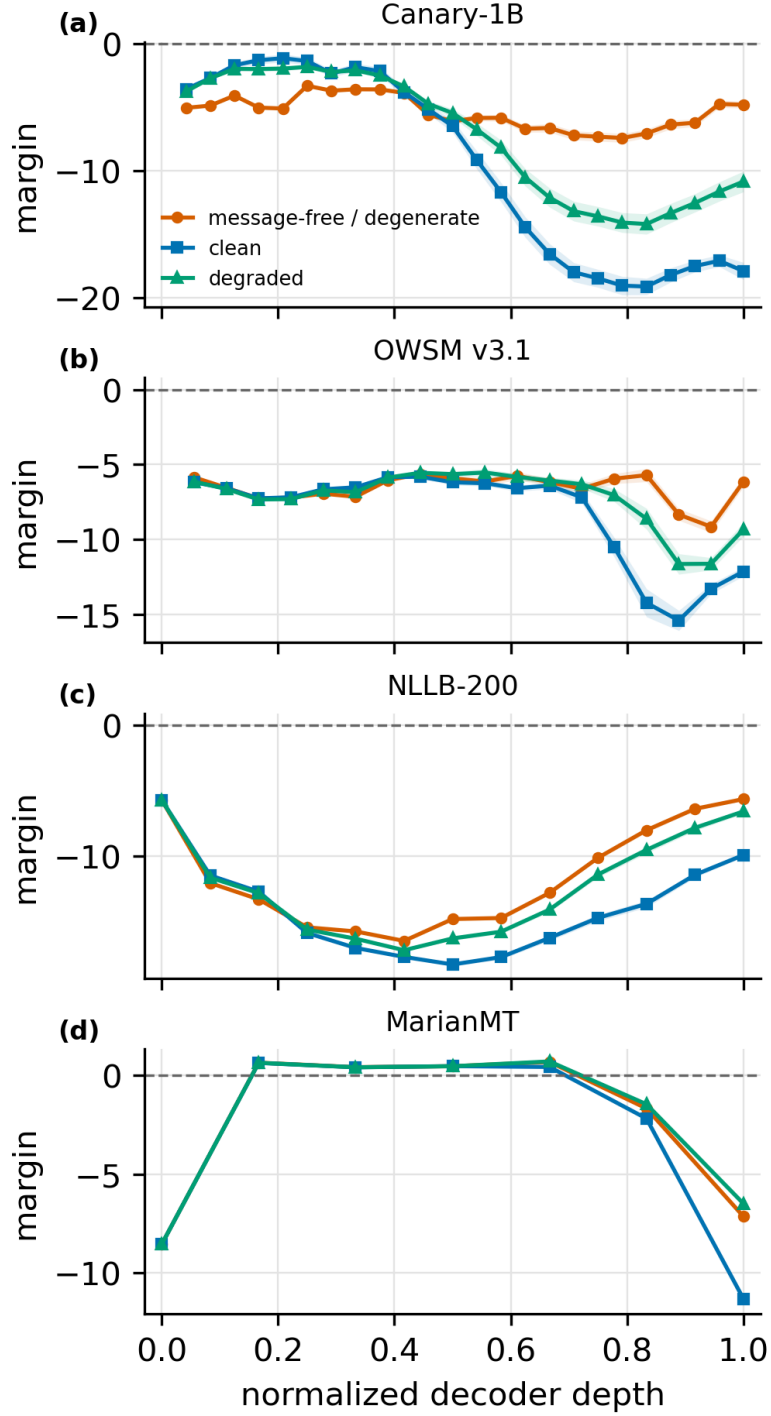


Figure 4: Cross-architecture logit-lens depth traces at decode step 0. Curves show the mean EOS abstain margin for message-free audio or degenerate source text, clean input, and degraded input; bands are normal-approximation 95% confidence intervals over $n = 120$ clips per condition for Canary/OWSM and $n = 200$ sources per condition for NLLB/Marian. The dashed line is the EOS argmax boundary. Message-free input finishes above clean input in every model, but below zero. Marian uniquely promotes EOS in the middle stack and demotes it in the final block, delimiting the stronger claim that abstention is never internally promoted.